

To the rescue, MacGyver

Inspired by the famous fictional secret agent Angus MacGyver, known for his troubleshooting skills, Georgia Tech submitted a research paper titled *Robots Using Environment Objects as Tools: The 'MacGyver' Paradigm for Mobile Manipulation* at the IEEE International Conference on Robotics and Automation (ICRA 2014). The MacGyver robots proposed in the paper stand out from the current generation in their skill to make use of environmental objects to solve problems, rather than be intimidated by unpredictable environments.

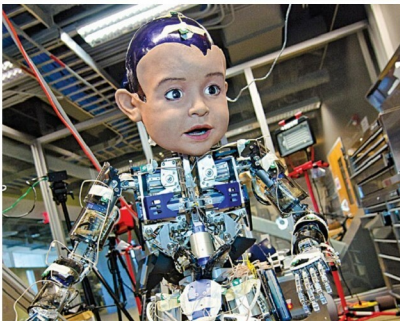
In a demonstration of the concept, the team designed a complete rescue scenario with a 100kg brick object blocking entry to a room and another 100kg loaded cart. Here is how Georgia Tech's Golem Krang robot used the myriad stuff around it to complete the task. They noted, "Interestingly, the loaded cart becomes a fulcrum for an arbitrary board to topple the bricks. Then the bricks, which were initially an obstacle, are used as a fulcrum for a lever to pry the door open. Finally, the robot uses a wider board to create a bridge and perform the simulated rescue."

Another team from Georgia Tech presented a second paper at the same conference detailing how a humanoid robot can be taught to traverse a gap by finding and using an object in its environment. This team achieves such behaviour using a concept that they call Environment Aware Planning (ETAP), which lets the robot evaluate and use resources in its environment to assist its locomotion capabilities, to achieve its goal.

In the demonstration, an HRP-2 robot was made to cross a rather large

gap. The robot, guided by an ETAP-based planning system, picked up a board in its environment and dropped it across the gap, resting on the two platforms on either side. To do this, it evaluated several boards and picked one that was larger than the gap to be crossed.

These researches show that it is



Diego-San, a boy-like humanoid teaching robot being developed at the University of California, San Diego (Courtesy: University of California, San Diego)

possible for robots of the future to understand their environment and react to it intelligently, enabling fully autonomous movement.

Teach me, crowd!

We have been speaking of robots that are aware of their environment and learn from it. But, that would mean their knowledge is only as good as what they see around them, and the people whom they learn from. If the behaviour or methods around them are flawed, their learning would be flawed too.

For example, if a robot is assigned the task of making something using building blocks, its construction would be only as good as the explanation or design it learns from the people around it, which may or may not be the best way of building the object. Instead, if a service like Amazon's Mechanical Turk could be used to

crowd-source designs for building the required object, and the robot could use machine learning to analyse these and choose the best way?

This was the technique that a team from the University of Washington proposed at a recent conference. They described their technique as a goal-based imitation learning framework that utilises crowd-sourcing as a major source of human demonstration data. They demonstrated their approach by making a robot build models with basic building blocks using knowledge gained from people around it as well as online crowd sources.

They also showed how such a technique could support human-robot collaboration tasks such as goal inference through object-part classification and missing-part prediction.

It is obvious that machine learning is making huge strides. Demonstrations of autonomous cars, intelligent industrial systems, smart drones, and even context-aware apps, are all signs of advancement of artificial intelligence—all of which is reflected in robotics too, albeit at much higher levels. There is no end to the examples we can give. Diego-San, a boy-like robot developed at the University of California, San Diego; user-friendly machine intelligence tools like PredictionIO that help develop self-learning robots and devices; Pepper, the chatty Japanese android that can understand people's emotions and around 70 to 80 per cent of their spontaneous conversation. The list will go on and on, till we arrive at the day when robots pass the Turing test with ease! ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai